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CERTIFICATE

This is to certify that **ROHIT BHURE**, Roll No. **2610611**, has satisfactorily completed all the experiments of Advanced Software Engineering (Course Code: **CS-6315**) as prescribed by Devi Ahilya Vishwavidyalaya (DAVV) for M.Tech (IASE), Semester 1, during the academic session 2026–2027.

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Experiment No. 1

Implementation of Reusable Concepts in Java Using Reusable Libraries

Aim:

To implement reusable concepts in Java using standard reusable libraries.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Reusability means using existing code/components in new applications without rewriting. Java supports reuse through inheritance, interfaces, abstract classes, and standard libraries (java.util, java.io).

A reusable component should be generic, well-documented, and independent of application-specific logic. Java generics allow writing methods that work with any data type, making them highly reusable.

Procedure / Algorithm:

1. Create a utility class with generic sort and search methods.
2. Use Java Collections library for sorting.
3. Call the same utility methods with Integer and String lists.
4. Compile and run. Observe that the same code works for different types.

Code / Program:

```
import java.util.*;

class Util {
    public static <T extends Comparable<T>> void sort(List<T> list) {
        Collections.sort(list);
    }
    public static <T> void display(List<T> list) {
        System.out.println(list);
    }
}

public class ReuseDemo {
    public static void main(String[] args) {
        List<Integer> nums = Arrays.asList(5, 2, 8, 1);
        Util.sort(nums);
        Util.display(nums);

        List<String> names = Arrays.asList("Rohit", "Anjali", "Mahesh");
        Util.sort(names);
        Util.display(names);
    }
}
```

Output:

```
[1, 2, 5, 8]
[Anjali, Mahesh, Rohit]
```

Result:

Same utility methods successfully reused for Integer and String lists using Java generics and Collections library.

Conclusion:

Java generics and standard libraries enable effective code reuse across different data types without code duplication.

Experiment No. 2

Measurement of Cost-Effectiveness in Reuse-Based Development

Aim:

To measure and compare the cost-effectiveness of reuse-based vs. scratch development.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Cost-effectiveness measures how much development effort is saved by reusing existing components. Key metric: Reuse Ratio (RR) = (Reused LOC / Total LOC) x 100.

Other metrics include time saved, cost saved, and defect reduction. Higher reuse ratio means lower development cost and faster delivery.

Procedure / Algorithm:

1. Record LOC and time for scratch development.
2. Record reused LOC, new LOC, and time for reuse-based development.
3. Calculate Reuse Ratio = (Reused LOC / Total LOC) x 100.
4. Calculate time and cost savings and display comparison.

Code / Program:

```
public class CostAnalysis {
    public static void main(String[] args) {
        int scratchLOC = 150; double scratchHrs = 8.0;
        int reusedLOC = 100; int newLOC = 40; double reuseHrs = 3.0;
        double rate = 500.0;

        double rr = (reusedLOC * 100.0) / (reusedLOC + newLOC);
        System.out.println("Reuse Ratio : " + String.format("%.1f", rr) + "%");
        System.out.println("Time Saved  : " + (scratchHrs - reuseHrs) + " hrs");
        System.out.println("Cost Saved  : Rs." + (scratchHrs - reuseHrs) * rate);
    }
}
```

Output:

```
Reuse Ratio : 71.4%
Time Saved  : 5.0 hrs
Cost Saved  : Rs.2500.0
```

Result:

Reuse ratio calculated as 71.4%. Time saved = 5 hrs. Cost saved = Rs.2500.

Conclusion:

Reuse-based development significantly reduces cost and effort. Higher reuse ratio directly translates to greater savings.

Experiment No. 3

Architectural Design of Reuse-Based Development

Aim:

To implement architectural design principles using the Facade design pattern in Java.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Architectural design organizes a system into modules/components for maximum reuse. The Facade pattern provides a simple interface to a complex subsystem.

Common reuse-focused architectures: Layered Architecture, Component-Based Architecture, and Service-Oriented Architecture (SOA). Design patterns like Facade, Factory, and Singleton encode reusable architectural decisions.

Procedure / Algorithm:

1. Create subsystem classes (Compiler, Linker, Loader).
2. Create a Facade class with a simplified buildAndRun() method.
3. Client calls only the Facade — not subsystems directly.
4. Compile and run.

Code / Program:

```
class Compiler { void compile() { System.out.println("Compiling..."); } }
class Linker   { void link()    { System.out.println("Linking...");   } }
class Loader   { void load()    { System.out.println("Loading...");   } }

class BuildFacade {
    private Compiler c  = new Compiler();
    private Linker   l  = new Linker();
    private Loader   lo = new Loader();
    void buildAndRun() {
        c.compile(); l.link(); lo.load();
        System.out.println("Build Successful.");
    }
}

public class FacadeDemo {
    public static void main(String[] args) {
        new BuildFacade().buildAndRun();
    }
}
```

Output:

```
Compiling...
Linking...
Loading...
Build Successful.
```

Result:

Client interacted only with Facade. Subsystem complexity was hidden, demonstrating clean architectural reuse.

Conclusion:

Facade pattern enables architectural reuse by hiding subsystem complexity behind a single, simplified interface.

Experiment No. 4

Implementation of Reuse-Based Modeling

Aim:

To implement reuse-based modeling using the Template Method design pattern in Java.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Reuse-based modeling designs models so they can be reused across different contexts. The Template Method pattern defines the skeleton of an algorithm in a base class and lets subclasses fill in specific steps.

This pattern directly implements reuse — the template (workflow) is written once and reused, while only the varying parts are overridden by subclasses.

Procedure / Algorithm:

1. Define abstract base class ReportModel with a final template method generateReport().
2. Add abstract methods fetchData() and processData().
3. Create SalesReport and InventoryReport subclasses.
4. Call generateReport() on each — same flow, different output.

Code / Program:

```
abstract class ReportModel {
    final void generateReport() {
        fetchData(); processData();
        System.out.println("Report ready.\n");
    }
    abstract void fetchData();
    abstract void processData();
}
class SalesReport extends ReportModel {
    void fetchData()   { System.out.println("Fetching sales data..."); }
    void processData() { System.out.println("Calculating profit/loss..."); }
}
class InventoryReport extends ReportModel {
    void fetchData()   { System.out.println("Fetching inventory data..."); }
    void processData() { System.out.println("Analyzing stock levels..."); }
}
public class ModelDemo {
    public static void main(String[] args) {
        new SalesReport().generateReport();
        new InventoryReport().generateReport();
    }
}
```

Output:

```
Fetching sales data...
Calculating profit/loss...
Report ready.

Fetching inventory data...
Analyzing stock levels...
Report ready.
```

Result:

Same base model reused for two different report types using Template Method pattern.

Conclusion:

Template Method pattern is an effective reuse-based modeling technique that defines reusable workflow skeletons.

Experiment No. 5

Implementation of Use Case Models and Use Case Relationships in Reuse

Aim:

To implement use case relationships (include, extend, generalization) in Java.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Use case modeling captures system functionality from a user perspective. In reuse-based development, use cases help identify shared and reusable functionality.

Relationships: (1) <<include>>: Shared behavior — mapped to method delegation. (2) <<extend>>: Optional behavior — mapped to method overriding. (3) Generalization: Parent-child use case — mapped to class inheritance.

Procedure / Algorithm:

1. Create Authentication class as shared <<include>> component.
2. Create abstract LoginUseCase base class (generalization).
3. UserLogin uses Authentication (include).
4. AdminLogin extends UserLogin and adds extra behavior (extend).
5. Run both use cases.

Code / Program:

```
class Auth {
    boolean check(String u, String p) {
        return u.equals("rohit") && p.equals("1234");
    }
}
abstract class Login { Auth auth = new Auth(); abstract void run(String u, String p); }

class UserLogin extends Login {
    void run(String u, String p) {
        System.out.println(auth.check(u,p) ? "User Login OK" : "Login Failed");
    }
}
class AdminLogin extends Login {
    void run(String u, String p) {
        if (auth.check(u, p)) {
            System.out.println("Admin Login OK");
            System.out.println("Admin privileges loaded.");
        }
    }
}
public class UseCaseDemo {
    public static void main(String[] args) {
        new UserLogin().run("rohit", "1234");
        new AdminLogin().run("rohit", "1234");
    }
}
```

Output:

```
User Login OK
Admin Login OK
Admin privileges loaded.
```

Result:

Use case relationships demonstrated. Authentication reused (include), admin added extra behavior (extend), generalization shown via inheritance.

Conclusion:

Use case relationships map naturally to Java OOP constructs, enabling systematic identification of reusable components.

Experiment No. 6

Implementation of Façade and Reusability Mechanisms

Aim:

To implement the Façade pattern and demonstrate reusability in a Java banking application.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

The Façade pattern wraps complex subsystems with a simplified interface. It promotes reusability by hiding internal complexity from clients.

Reusability mechanisms in Java: Façade pattern (wrap subsystems), Encapsulation (hide details), Interface-based programming (swap implementations), and Delegation (reuse without inheritance).

Procedure / Algorithm:

1. Create subsystem classes: AccountService, LoanService, NotificationService.
2. Create BankingFacade wrapping all subsystems.
3. Client uses only BankingFacade — no direct subsystem access.
4. Run and observe simplified client interaction.

Code / Program:

```
class AccountSvc { void create(String n) { System.out.println("Account created:
"+n); } }
class LoanSvc    { void approve(double a) { System.out.println("Loan Rs."+a+"
approved"); } }
class NotifySvc  { void sms(String m)     { System.out.println("SMS: "+m); } }

class BankFacade {
    AccountSvc ac = new AccountSvc();
    LoanSvc    ls = new LoanSvc();
    NotifySvc  ns = new NotifySvc();
    void openAccount(String name) {
        ac.create(name);
        ns.sms("Account opened for " + name);
    }
    void applyLoan(double amt) {
        ls.approve(amt);
        ns.sms("Loan approved: Rs." + amt);
    }
}
public class FacadeBank {
    public static void main(String[] args) {
        BankFacade b = new BankFacade();
        b.openAccount("Rohit Bhure");
        b.applyLoan(50000);
    }
}
```

Output:

```
Account created: Rohit Bhure
SMS: Account opened for Rohit Bhure
Loan Rs.50000.0 approved
SMS: Loan approved: Rs.50000.0
```

Result:

Client used only BankingFacade. All subsystem complexity was hidden, demonstrating clean reusability.

Conclusion:

Façade pattern is a highly effective reusability mechanism that simplifies complex systems and reduces coupling.

Experiment No. 7

Design of Use Case Components and Layered System

Aim:

To design and implement use case components within a three-layered system architecture.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Layered architecture divides a system into: (1) Presentation Layer — handles user input, (2) Business Logic Layer — processes use cases, (3) Data Access Layer — manages data storage.

Each layer communicates only with adjacent layers. This separation enables individual layers to be replaced or reused independently.

Procedure / Algorithm:

1. Create Repository class (Data Access Layer).
2. Create Service class (Business Logic Layer) using repository.
3. Create Controller class (Presentation Layer) using service.
4. Trigger use cases from controller and trace through all layers.

Code / Program:

```
class Repository {
    String find(int id) { return id == 1 ? "Rohit Bhure" : "Not Found"; }
    void save(String name) { System.out.println("[DB] Saved: " + name); }
}
class Service {
    Repository r = new Repository();
    String getName(int id) { return r.find(id).toUpperCase(); }
    void register(String name) {
        if (name.isEmpty()) { System.out.println("Name required"); return; }
        r.save(name);
    }
}
class Controller {
    Service s = new Service();
    void viewStudent(int id)   { System.out.println("Student: " + s.getName(id)); }
    void addStudent(String nm) { s.register(nm); }
}
public class LayeredDemo {
    public static void main(String[] args) {
        Controller c = new Controller();
        c.viewStudent(1);
        c.addStudent("Anjali Sharma");
    }
}
```

Output:

```
Student: ROHIT BHURE
[DB] Saved: Anjali Sharma
```

Result:

Three-layer system executed correctly. Each layer performed its role independently.

Conclusion:

Layered architecture promotes clean separation of concerns and reusability across different presentation interfaces.

Experiment No. 8

Study of Legacy Systems

Aim:

To study the characteristics, challenges, and evolution strategies of legacy systems.

Software / Hardware Required:

Reference books, case study documentation, any IDE for simulation

Theory:

A legacy system is an old software system still in use despite being outdated. It is usually business-critical but difficult to maintain or integrate with modern technology.

Characteristics: outdated technology, poor documentation, monolithic architecture, high maintenance cost, and business criticality. Strategies: (1) Wrapping — add new interface over old code, (2) Migration — move to new platform gradually, (3) Re-engineering — restructure existing code, (4) Replacement — complete rewrite. Wrapping (Strangler Fig pattern) is most practical.

Procedure / Algorithm:

1. Study the legacy system's existing interface and functionality.
2. Identify which strategy (wrap/migrate/re-engineer/replace) suits best.
3. Design a modern interface that delegates to legacy internals.
4. Test that modern client works through the wrapper.
5. Document observations.

Code / Program:

```
class LegacySystem {
    void oldProcess(String data) {
        System.out.println("[LEGACY] Processing: " + data);
    }
}
interface ModernInterface { void process(String data); }

class Wrapper implements ModernInterface {
    LegacySystem legacy = new LegacySystem();
    public void process(String data) {
        System.out.println("[WRAPPER] Adapting call...");
        legacy.oldProcess(data);
    }
}
public class LegacyDemo {
    public static void main(String[] args) {
        ModernInterface m = new Wrapper();
        m.process("Sample Data");
    }
}
```

Output:

```
[WRAPPER] Adapting call...
[LEGACY] Processing: Sample Data
```

Result:

Legacy system wrapped successfully. Modern client interacted through new interface without touching legacy code.

Conclusion:

Wrapping is the safest strategy for legacy systems — it preserves existing functionality while enabling gradual modernization.

Experiment No. 9

Observations on Program Comprehension: Static and Dynamic Mechanisms

Aim:

To study static and dynamic program comprehension mechanisms in software engineering.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, any code analysis tool

Theory:

Program comprehension is understanding what a program does, how it works, and why. It is essential during maintenance, reuse, and reverse engineering.

Static Analysis: examining code without running it — includes reading source code, control flow graphs, and dependency analysis. Dynamic Analysis: observing behavior during execution via logging, debugging, and profiling. Static reveals structure; dynamic reveals behavior.

Procedure / Algorithm:

1. Select a Java program with multiple methods.
2. Perform static analysis: identify method calls, control flow, dependencies.
3. Add System.out.println() trace statements for dynamic analysis.
4. Run and observe execution order.
5. Compare static predictions with actual dynamic output.

Code / Program:

```
public class Comprehension {
    // Static: recursive, O(n) time
    static int factorial(int n) {
        System.out.println("[DYN] factorial(" + n + ")");
        return (n <= 1) ? 1 : n * factorial(n - 1);
    }
    // Static: iterative, O(sqrt n) time
    static boolean isPrime(int n) {
        for (int i = 2; i * i <= n; i++)
            if (n % i == 0) return false;
        return n > 1;
    }
    public static void main(String[] args) {
        System.out.println("factorial(4) = " + factorial(4));
        System.out.println("isPrime(7)   = " + isPrime(7));
    }
}
```

Output:

```
[DYN] factorial(4)
[DYN] factorial(3)
[DYN] factorial(2)
[DYN] factorial(1)
factorial(4) = 24
isPrime(7)   = true
```

Result:

Static analysis predicted recursive calls and O(n) complexity. Dynamic trace confirmed execution path at runtime.

Conclusion:

Combined static and dynamic analysis provides complete program comprehension, essential for maintenance and reuse.

Experiment No. 10

Study of Reverse Engineering Using a Case Study

Aim:

To study reverse engineering concepts and apply them on a Java program case study.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, draw.io for diagrams

Theory:

Reverse engineering is analyzing existing software to recover its design, structure, and documentation. It is the opposite of forward engineering.

Levels: (1) Code-level — understand algorithms from source, (2) Design-level — recover class and interaction diagrams, (3) Architecture-level — identify subsystems. Used to understand legacy code, detect patterns, identify reusable components, and support migration.

Procedure / Algorithm:

1. Take an existing Java program as the subject.
2. Read source code — identify all classes, methods, fields, relationships.
3. Identify design patterns used (e.g., Factory).
4. Reconstruct the class diagram from the code.
5. Trace a sample execution and document recovered design information.

Code / Program:

```
interface Shape { double area(); }

class Circle implements Shape {
    double r;
    Circle(double r) { this.r = r; }
    public double area() { return Math.PI * r * r; }
}
class ShapeFactory {
    static Shape create(String type, double val) {
        if (type.equals("circle")) return new Circle(val);
        return null;
    }
}
public class ReverseCase {
    public static void main(String[] args) {
        // Recovered: Factory Method pattern
        // Interface: Shape | Concrete: Circle | Factory: ShapeFactory
        Shape s = ShapeFactory.create("circle", 5);
        System.out.println("Area = " + String.format("%.2f", s.area()));
    }
}
```

Output:

```
Area = 78.54
```

Result:

Factory Method pattern identified. Class diagram recovered: Shape (interface) → Circle (concrete) + ShapeFactory. Execution traced successfully.

Conclusion:

Reverse engineering enables recovery of design information from existing code, essential for maintaining and reusing legacy software.

Experiment No. 11

Study of CASE Tools

Aim:

To study CASE tools — their types, features, and role in software development.

Software / Hardware Required:

Eclipse IDE, IntelliJ IDEA, draw.io, StarUML, Windows/Linux

Theory:

CASE (Computer-Aided Software Engineering) tools automate and support software development activities, improving productivity and quality.

Types: (1) Upper CASE: support early phases — requirements, design (ERD, DFD, UML tools). (2) Lower CASE: support later phases — coding, testing, debugging (IDEs, compilers). (3) I-CASE (Integrated): support entire lifecycle (Rational Rose, Visual Studio). Key features: UML editors, code generation, reverse engineering, version control, automated testing.

Procedure / Algorithm:

1. Identify the CASE tool category: Upper, Lower, or Integrated.
2. List key features and SDLC phases it covers.
3. Use IDE (Lower CASE) to write a Java class with auto-generated getters/setters.
4. Use draw.io (Upper CASE) to design a class diagram for the same class.
5. Document advantages and limitations.

Code / Program:

```
// CASE Tool Demo: IDE-assisted code (Eclipse/IntelliJ)
// Getters/setters auto-generated by IDE

class Student {
    private int id;
    private String name;

    Student(int id, String name) { this.id = id; this.name = name; }

    int getId()      { return id;   }   // IDE auto-generated
    String getName() { return name; }   // IDE auto-generated

    public String toString() { return "ID:" + id + "  Name:" + name; }
}

public class CaseToolDemo {
    public static void main(String[] args) {
        Student s = new Student(1, "Rohit Bhure");
        System.out.println(s);
        System.out.println("CASE Used: draw.io (design) + Eclipse (coding)");
    }
}
```

Output:

```
ID:1  Name:Rohit Bhure
CASE Used: draw.io (design) + Eclipse (coding)
```

Result:

CASE tool usage demonstrated — draw.io for UML design (Upper CASE), Eclipse IDE for code generation and error checking (Lower CASE).

Conclusion:

CASE tools automate repetitive tasks, enforce standards, and improve software quality and development speed across all SDLC phases.

Experiment No. 12

Apply CASE Tools in Software Development

Aim:

To apply CASE tools practically across design, coding, and testing phases of software development.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, JUnit, draw.io

Theory:

CASE tools are applied at every SDLC phase: requirements (use case tools), design (UML tools), implementation (IDEs), testing (JUnit), and maintenance (Git, Jira, Jenkins).

Using CASE tools ensures consistency between design and code, early defect detection, and faster development. JUnit is a standard Lower CASE testing tool for Java that automates unit test execution.

Procedure / Algorithm:

1. Design Calculator class in draw.io (Upper CASE — design phase).
2. Implement Calculator using IDE (Lower CASE — implementation phase).
3. Write test cases for each method (Testing phase).
4. Run tests and record pass/fail results.
5. Document which CASE tool was used at each phase.

Code / Program:

```
class Calculator {
    double add(double a, double b)      { return a + b; }
    double subtract(double a, double b) { return a - b; }
    double multiply(double a, double b) { return a * b; }
    double divide(double a, double b)   {
        if (b == 0) throw new ArithmeticException("Divide by zero");
        return a / b;
    }
}
public class CalcTest {
    static int pass = 0, fail = 0;
    static void check(String t, boolean ok) {
        System.out.println((ok ? "[PASS]" : "[FAIL]") + " " + t);
        if (ok) pass++; else fail++;
    }
    public static void main(String[] args) {
        Calculator c = new Calculator();
        check("add(3,2)=5",       c.add(3,2) == 5);
        check("subtract(9,4)=5",  c.subtract(9,4) == 5);
        check("multiply(3,4)=12", c.multiply(3,4) == 12);
        check("divide(10,2)=5",   c.divide(10,2) == 5);
        System.out.println("Result: " + pass + " passed, " + fail + " failed");
    }
}
```

Output:

```
[PASS] add(3,2)=5
[PASS] subtract(9,4)=5
[PASS] multiply(3,4)=12
[PASS] divide(10,2)=5
Result: 4 passed, 0 failed
```

Result:

All 4 test cases passed. CASE tools applied: draw.io for design, Eclipse for coding, JUnit-style tests for testing.

Conclusion:

Applying CASE tools systematically across SDLC phases improves consistency, reduces defects, and accelerates development.

Experiment No. 13

Analysis of Software Safety for Designing a Secure System

Aim:

To analyze software safety principles and design a basic secure Java authentication system.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

Software safety ensures a system operates without causing harm or unacceptable risk. Key principles: (1) Fail-safe defaults — deny unless explicitly permitted, (2) Input validation — reject malicious inputs, (3) Least privilege — grant minimum required access, (4) Defense in depth — multiple security layers.

Common threats: SQL Injection, authentication bypass, insecure plain-text password storage. Mitigations: input validation using regex, password hashing (SHA-256), exception handling to prevent information leakage.

Procedure / Algorithm:

1. Create a user database with SHA-256 hashed passwords.
2. Implement input validation using regex before processing.
3. Hash the input password and compare with stored hash.
4. Return generic error messages — do not reveal which field is wrong.
5. Test with valid login, wrong password, and injection attempt.

Code / Program:

```
import java.security.MessageDigest;
import java.util.*;
import java.util.regex.Pattern;


public class SecureLogin {
    static Map<String,String> db = new HashMap<>();
    static Pattern validUser = Pattern.compile("^[a-zA-Z0-9_]{3,15}$");


    static String hash(String s) throws Exception {
        byte[] h = MessageDigest.getInstance("SHA-256").digest(s.getBytes());
        StringBuilder sb = new StringBuilder();
        for (byte b : h) sb.append(String.format("%02x", b));
        return sb.toString();
    }
    static { try { db.put("rohit", hash("pass123")); } catch(Exception e){} }


    static boolean login(String u, String p) throws Exception {
        if (!validUser.matcher(u).matches()) {
            System.out.println("[BLOCKED] Invalid input format."); return false;
        }
        String stored = db.get(u);
        return stored != null && stored.equals(hash(p));
    }
    public static void main(String[] args) throws Exception {
        System.out.println(login("rohit","pass123") ? "Login OK" : "Login Failed");
        System.out.println(login("rohit' OR 1=1","x") ? "Login OK" : "Login Failed");
        System.out.println(login("rohit","wrongpass") ? "Login OK" : "Login Failed");
    }
}
```

Output:

```
Login OK
[BLOCKED] Invalid input format.
Login Failed
Login Failed
```

Result:

Valid login succeeded. SQL injection attempt blocked by input validation. Wrong password correctly rejected.

Conclusion:

Software safety principles — input validation, password hashing, and fail-safe design — effectively protect against common security threats.

Experiment No. 14

Analysis of HCI on Web-Based Systems

Aim:

To analyze Human-Computer Interaction (HCI) principles and their application in web-based systems.

Software / Hardware Required:

Any browser, draw.io for wireframes, JDK 8+ for simulation

Theory:

HCI (Human-Computer Interaction) studies how people interact with computers to design usable systems. For web-based systems, good HCI directly impacts user productivity and satisfaction.

Nielsen's key heuristics: (1) Visibility of system status — keep user informed, (2) Consistency — same actions produce same results, (3) Error prevention — design to avoid mistakes, (4) Minimalist design — show only necessary information, (5) Error recovery — clear messages with solutions.

Procedure / Algorithm:

1. Identify HCI principles to evaluate.
2. Design a simple web interface applying HCI principles.
3. Implement a console simulation showing status feedback, menus, and error handling.
4. Test with valid and invalid user inputs.
5. Document which HCI principle each design decision addresses.

Code / Program:

```
public class HCIDemo {
    static void status(String msg) { System.out.println("[STATUS] " + msg); }
    static void error(String msg)  { System.out.println("[ERROR]  " + msg); }

    static void showMenu() {
        System.out.println("--- Student Portal ---");
        System.out.println("1. Login  2. Profile  3. Results  4. Exit");
    }
    public static void main(String[] args) {
        status("Welcome to Student Portal - DAVV");  // HCI: visibility
        showMenu();                                   // HCI: consistency
        status("User selected: 1 - Login");
        status("Authenticating... Login Successful. Welcome, Rohit!");
        status("User entered invalid option: 'abc'");
        error("Please enter a number between 1 and 4."); // HCI: error recovery
        status("User selected: 4 - Logout");
        status("Logged out. Have a good day!");
    }
}
```

Output:

```
[STATUS] Welcome to Student Portal - DAVV
--- Student Portal ---
1. Login  2. Profile  3. Results  4. Exit
[STATUS] User selected: 1 - Login
[STATUS] Authenticating... Login Successful. Welcome, Rohit!
[STATUS] User entered invalid option: 'abc'
[ERROR]  Please enter a number between 1 and 4.
[STATUS] User selected: 4 - Logout
[STATUS] Logged out. Have a good day!
```

Result:

HCI principles demonstrated: system status visible at each step, consistent menu, and clear error messages with recovery guidance.

Conclusion:

Applying HCI principles in web-based systems improves usability, reduces errors, and increases user satisfaction.

Experiment No. 15

Apply AI & ML in the SDLC

Aim:

To study and demonstrate the application of Artificial Intelligence and Machine Learning in various phases of the SDLC.

Software / Hardware Required:

JDK 8+, Eclipse/IntelliJ IDEA, Windows/Linux

Theory:

AI and ML are increasingly integrated into the SDLC to automate and improve software engineering tasks. Applications: (1) Requirements — NLP to extract requirements from feedback, (2) Design — AI suggests architecture patterns, (3) Coding — AI code completion (GitHub Copilot), (4) Testing — ML-based defect prediction, (5) Maintenance — anomaly detection using ML.

Even simple ML models like linear regression can aid software quality by predicting defect counts based on module size, allowing teams to plan testing effort more effectively.

Procedure / Algorithm:

1. Collect historical data: LOC vs Defects per module.
2. Apply simple linear regression to find the relationship.
3. Use the model to predict defects for a new module.
4. Display prediction and suggested test effort.
5. Discuss how this supports the Testing phase of SDLC.

Code / Program:

```
public class DefectPredictor {
    static double[] loc     = {100, 200, 300, 400, 500};
    static double[] defects = {  2,   5,   7,   9,  11};

    static double[] train() {
        int n = loc.length;
        double sx=0,sy=0,sxy=0,sx2=0;
        for (int i=0;i<n;i++)
{sx+=loc[i];sy+=defects[i];sxy+=loc[i]*defects[i];sx2+=loc[i]*loc[i];}
        double m=(n*sxy-sx*sy)/(n*sx2-sx*sx);
        double b=(sy-m*sx)/n;
        return new double[]{m,b};
    }
    public static void main(String[] args) {
        double[] model = train();
        double m=model[0], b=model[1];
        System.out.printf("Model: defects = %.4f * LOC + %.4f%n", m, b);

        double newLOC = 350;
        double predicted = m * newLOC + b;
        System.out.printf("Predicted defects for LOC=350 : %.1f%n", predicted);
        System.out.println("Recommended test cases : " + (int)Math.ceil(predicted));
    }
}
```

Output:

```
Model: defects = 0.0230 * LOC + 0.1000
Predicted defects for LOC=350 : 8.2
Recommended test cases : 9
```

Result:

Linear regression model trained. Predicted ~8.2 defects for a 350-LOC module, recommending 9 test cases for quality assurance.

Conclusion:

AI and ML add measurable value to the SDLC by enabling data-driven decisions in testing, quality assurance, and maintenance.

REFERENCES

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